# RADBox Equipment Startup Guide

RADBox (Real-time Air Quality Data Box) is a comprehensive air quality monitoring system designed under a philosophy of low cost, ease of use, and open source. Its fundamental purpose is to facilitate access for students and citizens to understand and measure atmospheric pollutants in real time, helping them become more connected to environmental science. The system is based on a DIY (Do-It-Yourself) architecture that integrates microcontrollers such as the Arduino Nano Every and low-cost specialized sensors, making the technology accessible without the need for complex laboratory equipment.

Specifically, RADBox uses the Grove SCD41 and Grove SGP41 sensors, which are responsible for measuring critical environmental variables such as $CO_2$, temperature, humidity, and the presence of VOC (volatile organic compounds) and $NO_x$ (nitrogen oxides) gases.

The Arduino microcontroller acts as the system's communication bridge, collecting these electrical signals and sending them via a USB cable through a serial port to a computer. Once the information reaches the computer, a script developed in Python handles the final stage of the process. This program converts the raw electrical signals into readable data and standard units of measurement, enabling the generation of real-time graphs using specialized libraries. This immediate visualization is key for users to observe how different external conditions affect air quality.

Finally, all information collected during monitoring sessions is automatically stored in a .csv format file. This backup allows the data not only to be seen in the moment but also to be analyzed in greater depth later using common tools such as Excel. In this way, RADBox fulfills its mission of demonstrating that it is possible to manufacture functional and economical scientific instruments, transforming abstract concepts about pollution into tangible and understandable data that foster scientific knowledge and ecological awareness in any educational environment.

# Guide for Equipment Configuration and Startup

To get the equipment up and running, it is necessary to do the following:

## Step 0: Assemble RADBox and configure the Arduino to start its operation

1. If you haven't already built RADBox equipment, it will be necessary to assemble and configure it beforehand. To do so, go to the repository using the following link: https://github.com/Sofializarazo/RADbox/tree/Radbox_proyect
2. Inside the "instructions" folder, you will find a file called "Setup Instructions", which contains detailed instructions for the assembly and initial configuration of the equipment.

If you already have the equipment assembled and the Arduino configured with the code, then proceed to step 1.

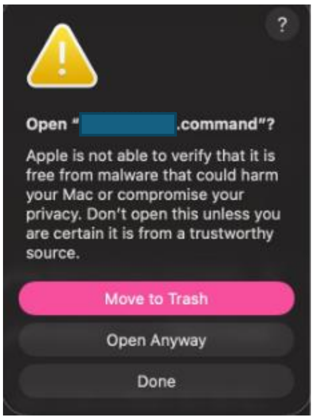
**Important:** Please read the instructions carefully before starting; this will prevent you from making mistakes while following them.

## Step 1: Downloading the Base Software

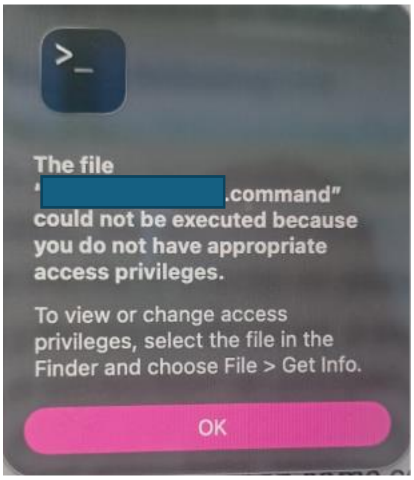
The software required to view the data generated by the device is hosted in a GitHub repository. Follow these instructions to obtain it:

1. Access the repository by entering the following link: https://github.com/Sofializarazo/RADbox/tree/Radbox_proyect
2. If you are in **Windows**, go to the folder named “app” and download the file “RADBox_lite.exe”or if you are in **MacBook** download the file “RADBox_litemac.command”.
3. Once downloaded, save in a folder on your desktop for easy access.
    - If working on **Windows** a security message appears while downloading the file, indicating that the file comes from an unknown source or asking whether you want to continue with the download, select “Yes” or “Continue anyway.” The file is safe. This warning may appear on some computers because the system does not recognize the source of the file and, for security reasons, alerts the user before opening it. In this case, simply confirm the download and continue the installation process.
    - If, when trying to open the application on **MacBook**, a security message appears indicating that the file comes from an unknown source and cannot be opened securely because it cannot be verified for malware, do not worry. This is an additional security measure implemented by macOS. To resolve it, go to the Apple menu in the top-left corner, open “System

Settings”, access “Privacy & Security”, and scroll down to the Security section on the right panel. There, you will see a message similar to: “RADBoxmac.command was blocked because it is not from an identified developer.” In that case, click “Open Anyway”,enter your password, give the permissions and the application will recognize it as safe in future launches.




If it still won't open and now when you try to open it the message that appears is the following:




This is the solution: Press Command + Space to open Spotlight. Type “Terminal” and press Enter. Write “chmod +x ” (Important: a space must be left between the x and the next part of the instruction.) Drag and drop the RADBoxmac.command file directly into the Terminal window and then press enter.

If you have an Intel-based **MacBook**, the program should run without any issues. However, if your Mac has an Apple Silicon processor, you must complete one final step before running the software. If you are unsure which processor your Mac has, simply open the RADBoxmac.command file to check the Terminal window output. If it displays “zsh: bad CPU type in executable...” followed by “[process completed]”, your Mac has an Apple Silicon processor and requires the additional step; if it takes a moment to load and says “Please plug in the Arduino”, your Mac has an

Intel processor, meaning no further action is required and you can proceed to the next step. If the output indicates you have an Apple Silicon Mac, go back to the Terminal window from the previous step, type “softwareupdate --install-rosetta --agree-to-license", and press Enter. Once the installation is complete, the application will be ready to use, which you can confirm by opening the app and waiting about a minute until the message “Please plug in the Arduino” appears, indicating the software is running.

- Do not change the name of the source file, as this could cause errors.

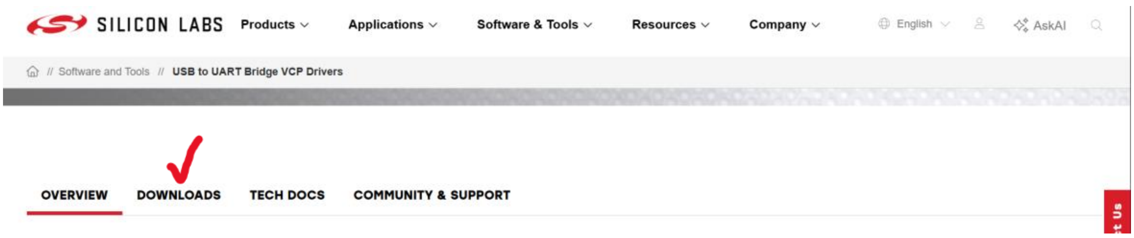
4. If you need to make modifications to the software, please refer to Appendix 1.

## Step 2: Driver Installation (USB to UART)

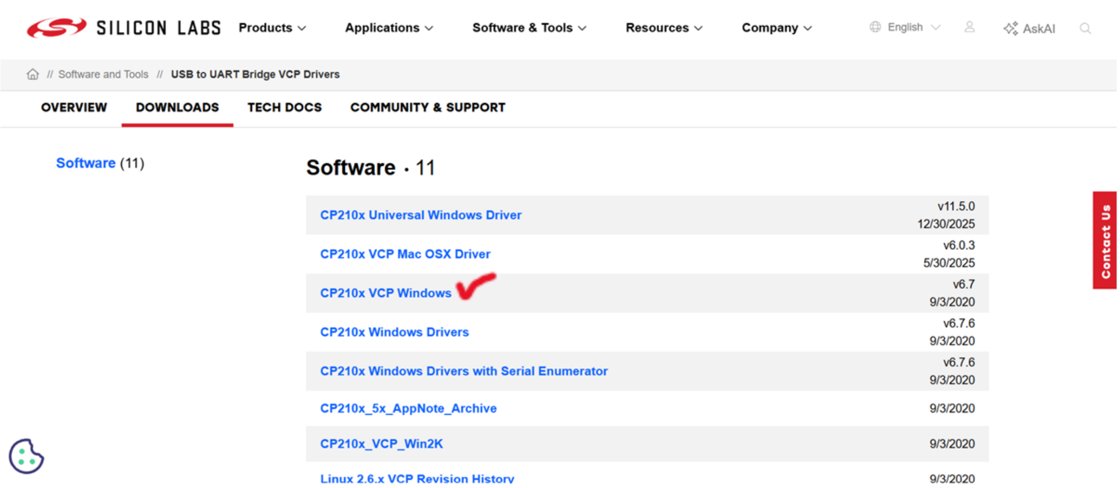
For the computer to recognize the equipment when connected, you must install the driver that enables serial communication.

### On Windows:

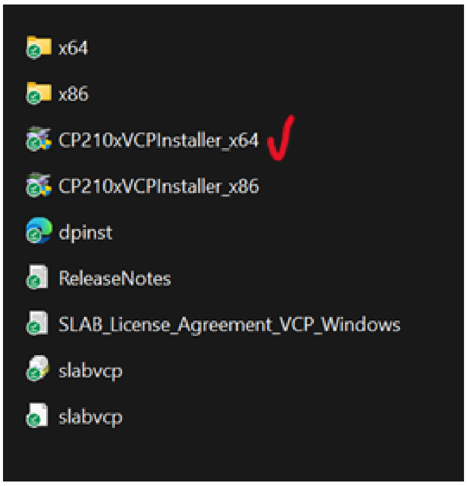
1. Access the Silicon Labs download portal through the following link: [CP210x USB to UART Bridge VCP Drivers.]()
2. Click on the "Downloads" tab.




3. In this section, you will see a panel of options such as the one shown below:



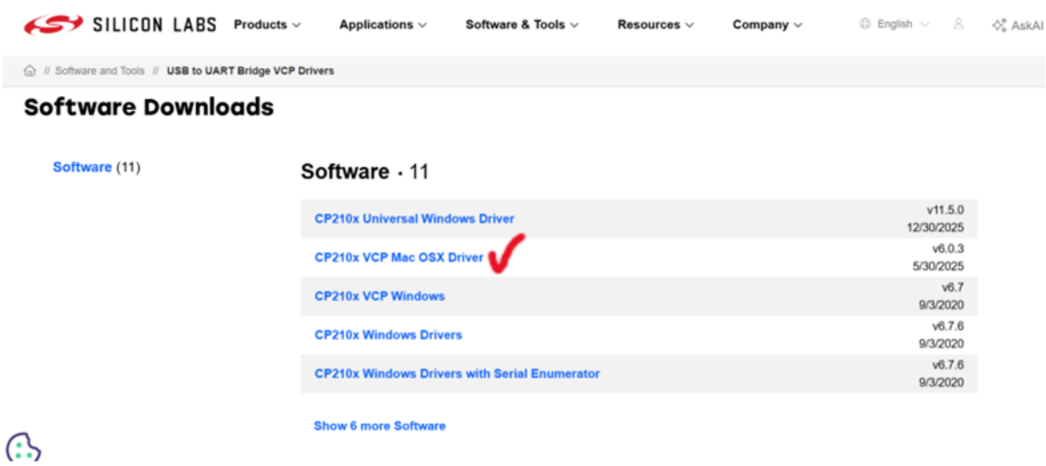

4. Locate and click on the file named "CP210x VCP Windows". This step will begin the download of a compressed folder in .zip format.
5. Unzip the folder to your desktop to make managing the files easier.
6. Open the unzipped folder and run the file "CP210xVCPInstaller_x64" by double-clicking it. Follow the on-screen instructions and authorize the system permissions to complete the installation.



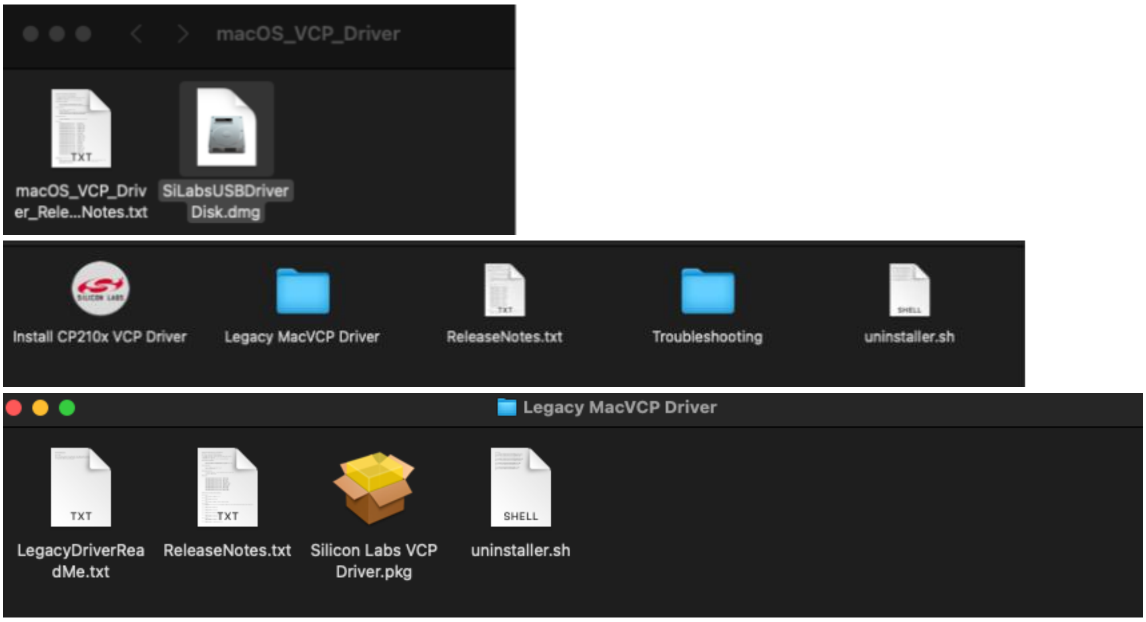

## On MacBook:

1. On most Macs, this step isn't necessary, so you can skip directly to "Step 3: System Connection and Execution". If it doesn't work, come back here and proceed with the standard installation.
2. Access the Silicon Labs download portal through the following link: [CP210x USB to UART Bridge VCP Drivers.]()
3. Click on the "Downloads" tab.


4. In this section, you will see a panel of options such as the one shown below:




5. Locate and click on the file named "CP210x VCP Mac OSX Driver". This step will begin the download of a compressed folder in .zip format.
6. Unzip the folder to your desktop to make managing the files easier.
7. In several of the following steps, you may be asked for your computer password to continue with the installation. In those cases, grant the necessary permissions by entering your password when prompted, and likewise click on the "Continue" button or its equivalent whenever the system indicates it.
8. When you download the file, you will find two documents inside the folder. Double-click the file named "SilLabsUSBDriverDisk.dmg", then, within it, open the "Legacy MacVCP Driver" folder, and subsequently double-click the "Silicon Labs VCP Driver.pkg" file. Next, grant all requested permissions to complete the installation of the program.

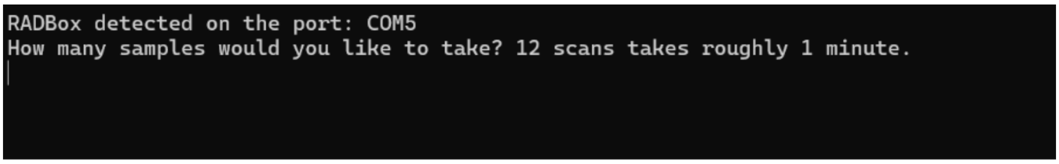



9. If, at the end of the installation process, a message appears indicating that the software is outdated, do not worry. Simply close the window by clicking the “X” or “Cancel” and continue with the process, without deleting or moving the file to the trash.

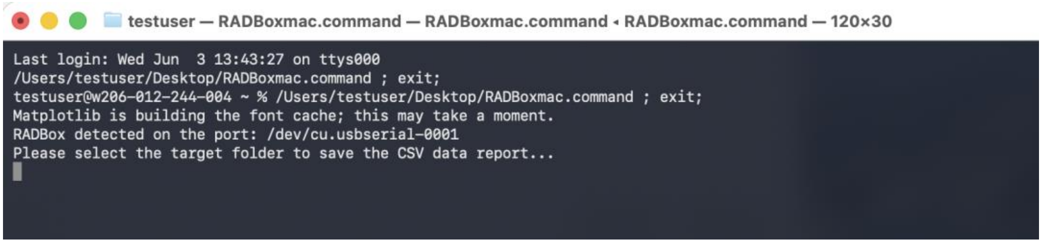
## Step 3: System Connection and Execution

Once the drivers are installed and the base software is downloaded, the system is ready to operate:

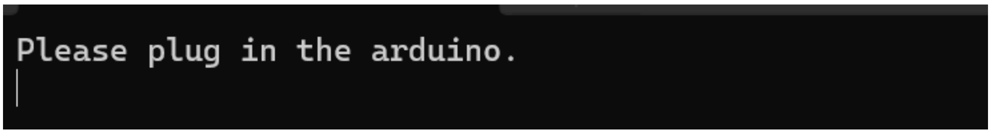
1. Connect the RADBox equipment to an available USB port on your computer.
2. Navigate to the folder where you saved the RADBox file and launch the main program by double-clicking the executable file.
3. After opening the .exe file, a command window or configuration interface will appear that looks like the following (It takes approximately one to one and a half minutes to appear):
   - For **Windows,** the data will be automatically saved in the folder where the application is located. Simply enter the number of data points you wish to collect and press Enter.




   - For **Macbook**, a more extensive warning message may appear. The system may indicate that Matplotlib is building the front cache; this may take a moment; in that case, ignore the message, as the application is starting correctly. Once the application loads properly, you will be asked to select the folder where you want to save the data. Choose the location that is most convenient for you. Afterwards, the system will ask how many samples you wish to take. Enter the corresponding number and press Enter. At that point, data acquisition will begin automatically.

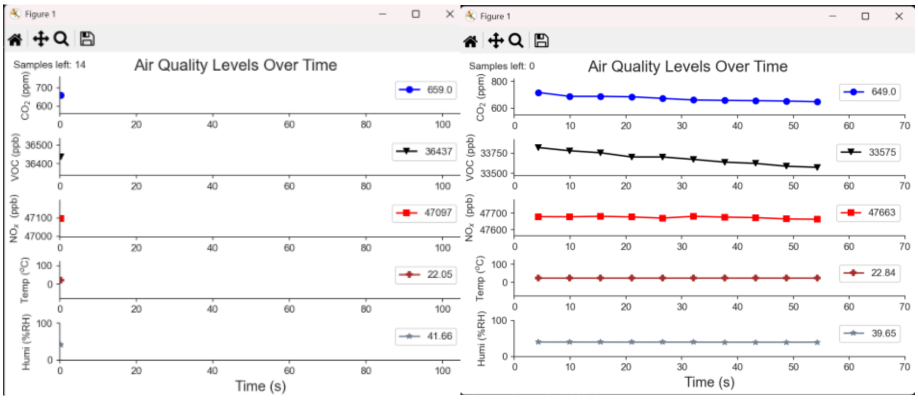



If you are disconnected, the following message will appear:

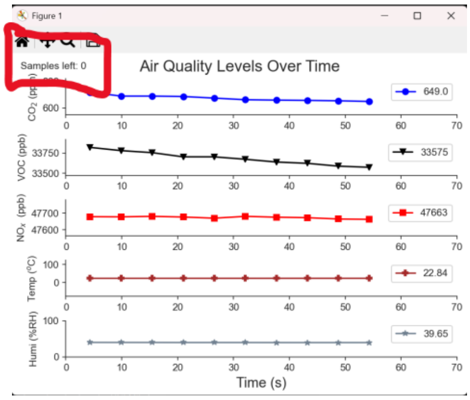



Unplug it and plug it back in or plug it into a different port.

4. Manually enter the amount of data you wish to collect, type the number, and press the Enter key.
5. The system will start the capture automatically, so don't close the application until it has finished running. The data will be plotted showing trends and variations over time.




6. Once the sampling is complete, the program will display "Samples left:0" in the top left corner.




7. You can safely close the application; no prior manual saving is necessary.

## Step 4: Storage of Results

1. The data is automatically stored in the same folder where the executable file is located for **Windows**, or in the folder selected during data acquisition for **MacBook**. It is recommended to rename the generated file immediately after its creation to facilitate identification and future analysis.
2. Data is automatically exported in .csv format (comma-separated values). To consult them after disconnecting the RADBox, simply double-click the generated file to open it with Microsoft Excel.
3. Generally, this is not necessary; however, if the data does not appear in columns when you open the file, please follow these steps:
    - Go to this link: https://convertio.co/csv-xlsx/
    - Select your file and click Convert to .xlsx format.
    - Once the process is finished, download the file. When opened in Excel, the data will be correctly organized into columns.

4. Upon opening the file, the data will be presented in organized columns, ready to be processed or analyzed.

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | Time (s) | CO2 (ppm | VOC (ppb) | NOx (ppb) | Temperat | Humidity (%RH) | |
| 2 | 4.33 | 718 | 33821 | 47679 | 22.74 | 40.11 | |
| 3 | 9.88 | 689 | 33781 | 47678 | 22.75 | 40 | |
| 4 | 15.44 | 689 | 33756 | 47681 | 22.77 | 39.94 | |
| 5 | 21 | 686 | 33704 | 47677 | 22.78 | 39.79 | |
| 6 | 26.55 | 674 | 33704 | 47670 | 22.79 | 39.82 | |
| 7 | 32.1 | 662 | 33674 | 47681 | 22.78 | 39.79 | |
| 8 | 37.66 | 659 | 33640 | 47676 | 22.81 | 39.69 | |
| 9 | 43.21 | 657 | 33624 | 47673 | 22.83 | 39.63 | |
| 10 | 48.76 | 654 | 33589 | 47665 | 22.84 | 39.62 | |
| 11 | 54.31 | 649 | 33575 | 47663 | 22.84 | 39.65 | |
| 12 | | | | | | | |
| 13 | | | | | | | |
| 14 | | | | | | | |

5. To recreate the visual trends within Excel, follow the procedure mentioned below:
   - Highlight the data columns you wish to graph.

| Time (s) | CO2 (ppm | VOC (ppb | NOx (ppb | Temperat | Humidity (%RH) |
|---|---|---|---|---|---|
| 4.33 | 718 | 33821 | 47679 | 22.74 | 40.11 |
| 9.88 | 689 | 33781 | 47678 | 22.75 | 40 |
| 15.44 | 689 | 33756 | 47681 | 22.77 | 39.94 |
| 21 | 686 | 33704 | 47677 | 22.78 | 39.79 |
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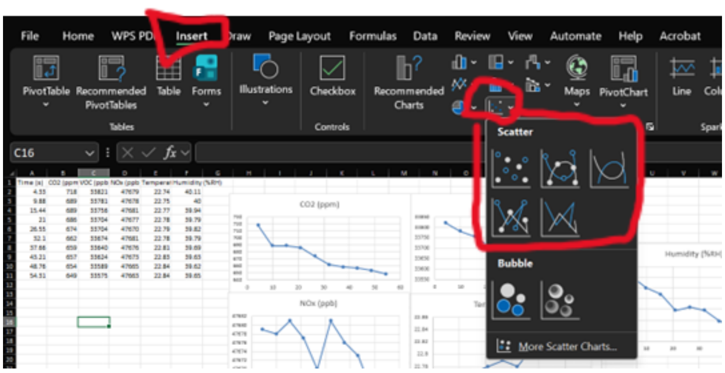
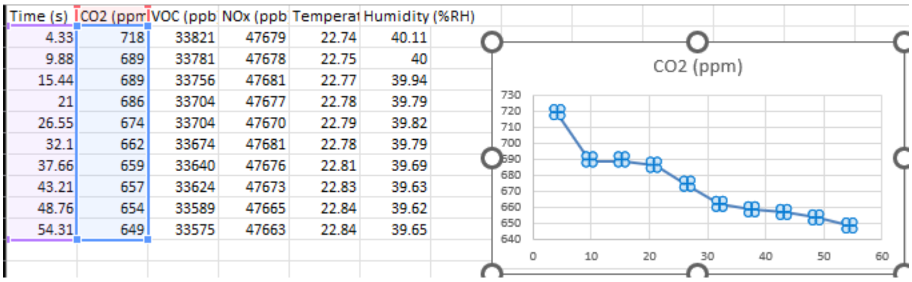
| 37.66 | 659 | 33640 | 47676 | 22.81 | 39.69 |
| 43.21 | 657 | 33624 | 47673 | 22.83 | 39.63 |
| 48.76 | 654 | 33589 | 47665 | 22.84 | 39.62 |
| 54.31 | 649 | 33575 | 47663 | 22.84 | 39.65 |

   - Go to the Insert tab in the top.
   - Click on the Scatter icon:  This will generate a graph where each point represents an individual reading, allowing you to observe the data distribution.
   - If you prefer to visualize the progression using lines to better identify the behavior of the variables, select the chart options with trend lines:



- If you accidentally selected the wrong data, click on one of the points in the graph and you will see the selected columns highlight and change color. You can then click on them and move them to wherever you want.



| Time (s) | CO2 (ppm | VOC (ppb | NOx (ppb | Temperat | Humidity (%RH) |
|---|---|---|---|---|---|
| 4.33 | 718 | 33821 | 47679 | 22.74 | 40.11 |
| 9.88 | 689 | 33781 | 47678 | 22.75 | 40 |
| 15.44 | 689 | 33756 | 47681 | 22.77 | 39.94 |
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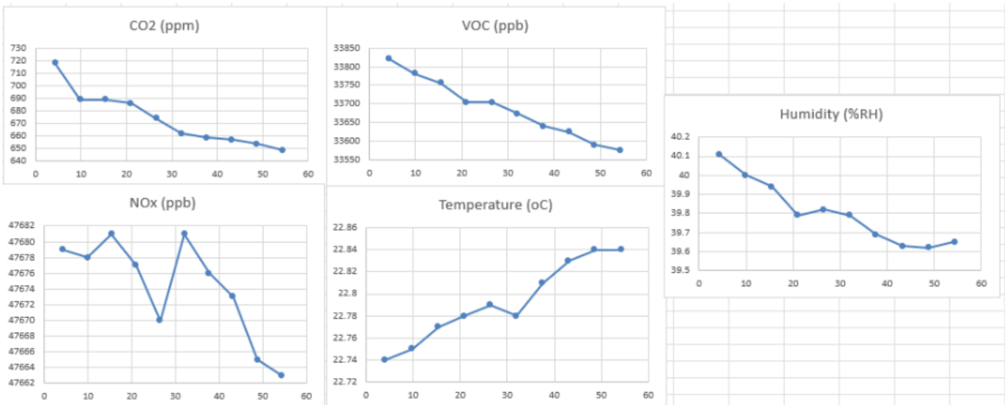
| 48.76 | 654 | 33589 | 47665 | 22.84 | 39.62 |
| 54.31 | 649 | 33575 | 47663 | 22.84 | 39.65 |



- The data graphed from Excel will look like this:




## Appendix 1: Guide for Python Code Editing

This Appendix is designed for users who wish to explore the internal functioning of the software, add custom functions, or run the program directly from the source code.

The **.**py file (source code) is located in the same GitHub link within the source_code folder. Navigate to that folder and download the RADBox.py file for **Windows** or RADBoxmac.py for **MacBook**.

To view and edit it correctly, follow these steps:

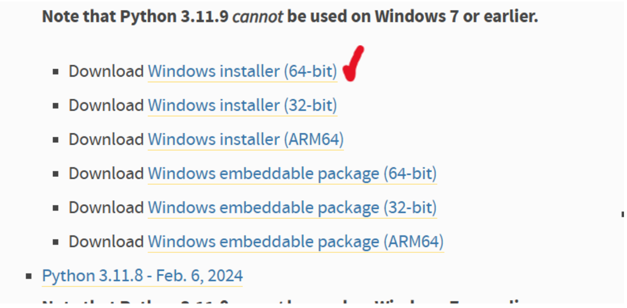
## On Windows:

### 1. Development Environment Setup

Although multiple text editors exist, for this manual we will use Visual Studio Code (VS Code) for its versatility and integrated tools.

1. Download and install the editor from the official site: [code.visualstudio.com](https://code.visualstudio.com) .
2. To ensure code compatibility, Python version 3.11.9 is required; therefore:

a. Go to: python.org/downloads/windows.
b. Locate and download the installer: "Download Windows installer (64-bit)"

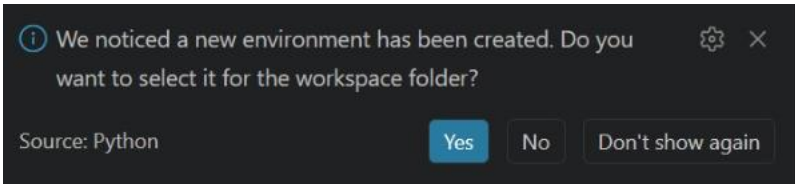



c. During installation, make sure to check the box "Add Python to PATH" before completing the process; if this option is not shown, simply accept all permissions.

## 2. Visual Studio Code Preparation

Once both programs are installed, configure the editor as follows:

1. Copy your .py file and paste it into a new folder created exclusively for these project modifications.
2. Open Visual Studio Code, go to the top menu, and select File > Open Folder.
3. Search for and select the folder you just created, then click Select Folder.
4. In the left panel (Explorer), click on your file to open it in the editor.
5. To interact with the language, you must enable the integrated terminal; to do this, go to the top menu and select View > Terminal.
6. In the terminal that appears at the bottom, you can confirm that the system recognizes the Python installation correctly; to do this, type the following command and press Enter: "python –version". You should see a message confirming the version: Python 3.11.9.
7. To avoid conflicts with other projects, create an isolated environment with the specific version of Python by typing: "py -3.11 -m venv Radbox".
8. If a notification appears in the bottom right corner asking if you want to use the new environment for the folder, select Yes.




9. Execute the following command to start working within the project environment: ".\Radbox\Scripts\activate"
10. Confirm that the environment is active when the text (Radbox) appears in green at the beginning of the command line.

## 3. Library Installation

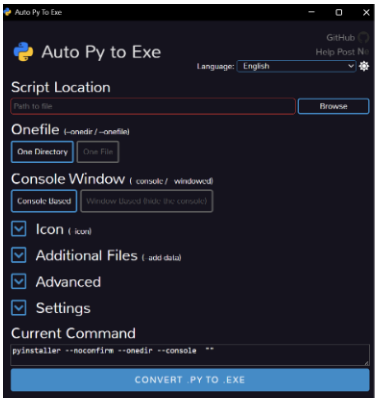
For the code to function correctly, it is necessary to install the libraries that manage communication with the equipment and the generation of graphs.

1. Copy and paste the following command to install the necessary libraries: “py -m pip install pyserial matplotlib seaborn”
2. To verify that the process was successful, type: “py -m pip list”. Ensure that pyserial (v3.5 or higher), matplotlib (v3.6.3 or higher), and seaborn (v0.12.1 or higher) appear in the list.
3. The code is ready to be edited to your liking.

## 4. Converting Code (.py) to Executable (.exe)

Once you have made the desired edits to the code, you can convert it into a standalone application to run it without the need to open the code editor.

1. In the terminal write “py -m pip install auto-py-to-exe” and press Enter.
2. Once the download is finished, type the following command in the same window: “py -m auto_py_to_exe”
3. A graphical interface will open.




Configure the fields as follows:

a. Script Location: Click on “Browse” and select your file named RADBox.py or the name you have given it.
b. Onefile: Select the “One File” option to generate a single executable file.
c. Console Window: Select “Console Based”.
d. Settings: In “Output Directory”, define a new folder where you want to save your final application; your data will also be saved in this same folder.

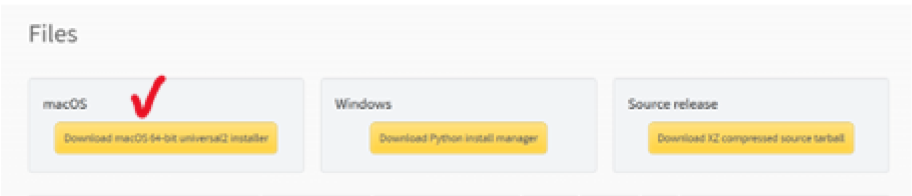
4. Click the blue button Convert .py to .exe. Once the process is finished, your new executable file will be ready to operate with the RADBox equipment.

## On MacBook:

### 1. Development Environment Setup

Although multiple text editors exist, for this manual we will use Visual Studio Code (VS Code) for its versatility and integrated tools.

1. Download and install the editor from the official site: https://code.visualstudio.com/Download Download for Mac
2. To ensure code compatibility, Python version 3.11.9 is required; therefore:
   a. Go to: https://www.python.org/downloads/release/python-3119/
   b. Locate and download the installer: “Download macOS 64-bit universal2 installer”




   c. During installation, make sure to check the box "Add Python to PATH" before completing the process; if this option is not shown, simply accept all permissions.

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5. To interact with the language, you must enable the integrated terminal; to do this,go to the top menu and select View > Terminal.
6. In the terminal that appears at the bottom, you can confirm that the system recognizes the Python installation correctly; to do this, type the following command and press Enter: “python3.11 --version”. You should see a message confirming the version: Python 3.11.9.

7. To avoid conflicts with other projects, create an isolated environment with the specific version of Python by typing: “python3.11 -m venv Radbox”.
8. Execute the following command to start working within the project environment: “source Radbox/bin/activate”
9. Confirm that the environment is active when the text (Radbox) appears at the beginning of the command line.

## 3. Library Installation

For the code to function correctly, it is necessary to install the libraries that manage communication with the equipment and the generation of graphs.

1. Copy and paste the following command to install the necessary libraries: “pip install pyserial matplotlib seaborn”
2. The code is ready to be edited to your liking.
3. To run the code, type “python3 RADBoxmac.py”

## 4. Converting Code (.py) to Executable

Once you have made the desired edits to the code, you can convert it into a standalone application to run it without the need to open the code editor.

1. In the terminal, type the following: “pip install pyinstaller”
2. Now enter: “pyinstaller --noconfirm --onefile --console --name="RADBoxmac" RADBoxmac.py”
3. Then, “mv dist/RADBoxmac dist/RADBoxmac.command && chmod +x dist/RADBoxmac.command”
4. And that’s it; the executable file will be saved in the folder inside another folder called dist.